Probleme examen AA

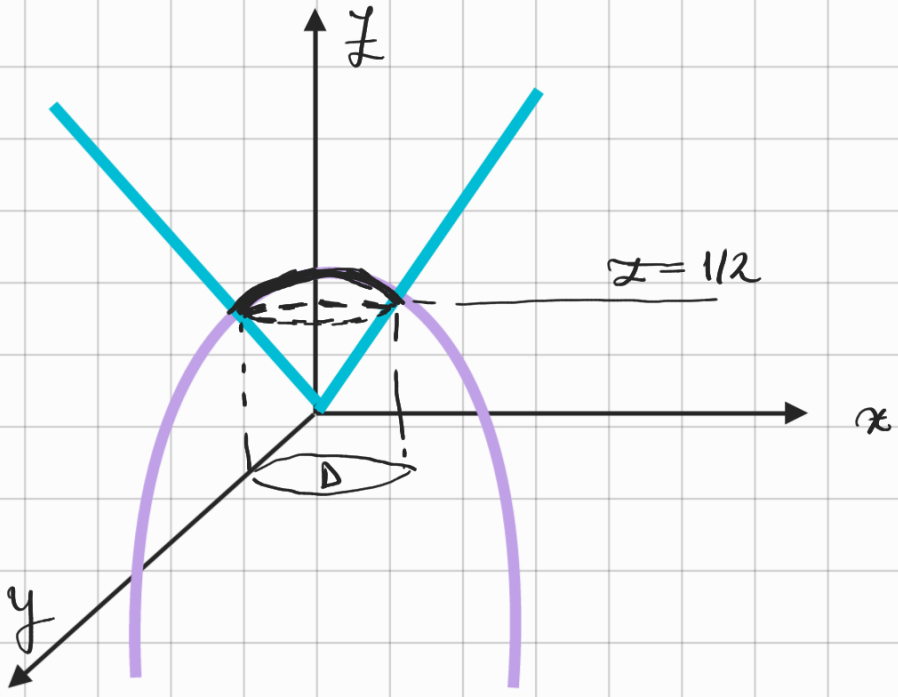


Ex. cu aria

→ paraboloid

Să se determine aria porțiunii de pe suprafața $1 - z = x^2 + y^2$, cu $z > 0$, situată în interiorul suprafeței $x^2 + y^2 = 2z^2$. Reprezentare grafică.

↳ con



$1 - z = 2z^2 \Leftrightarrow$

$2z^2 + z - 1 = 0$

$\Delta = 1 + 8 = 9$

$z_{1,2} = \begin{cases} -1 \\ 1/2 \end{cases}$, dar $z > 0$

$z = f(x,y) = 1 - x^2 - y^2$

$$\left.\begin{aligned} \frac{\partial f}{\partial x} &= -2x \\ \frac{\partial f}{\partial y} &= -2y \end{aligned}\right\} \Rightarrow$$

aria suprafeței este $\mathcal{A} = \int_{\Sigma} d\sigma = \iint_D \sqrt{1 + 4x^2 + 4y^2}\, dx\, dy$

unde $D = \{(x,y) \mid x^2 + y^2 \le \frac{1}{2}\} \Rightarrow$

Jacobian: ρ

Parametrizare D: $\begin{cases} x = \rho \cos t \\ y = \rho \sin t \end{cases}$ $\quad \rho^2 \le \frac{1}{2} \Rightarrow$ $\begin{cases} \rho \in [0, \frac{1}{\sqrt{2}}] \\ t \in [0, 2\pi] \end{cases}$

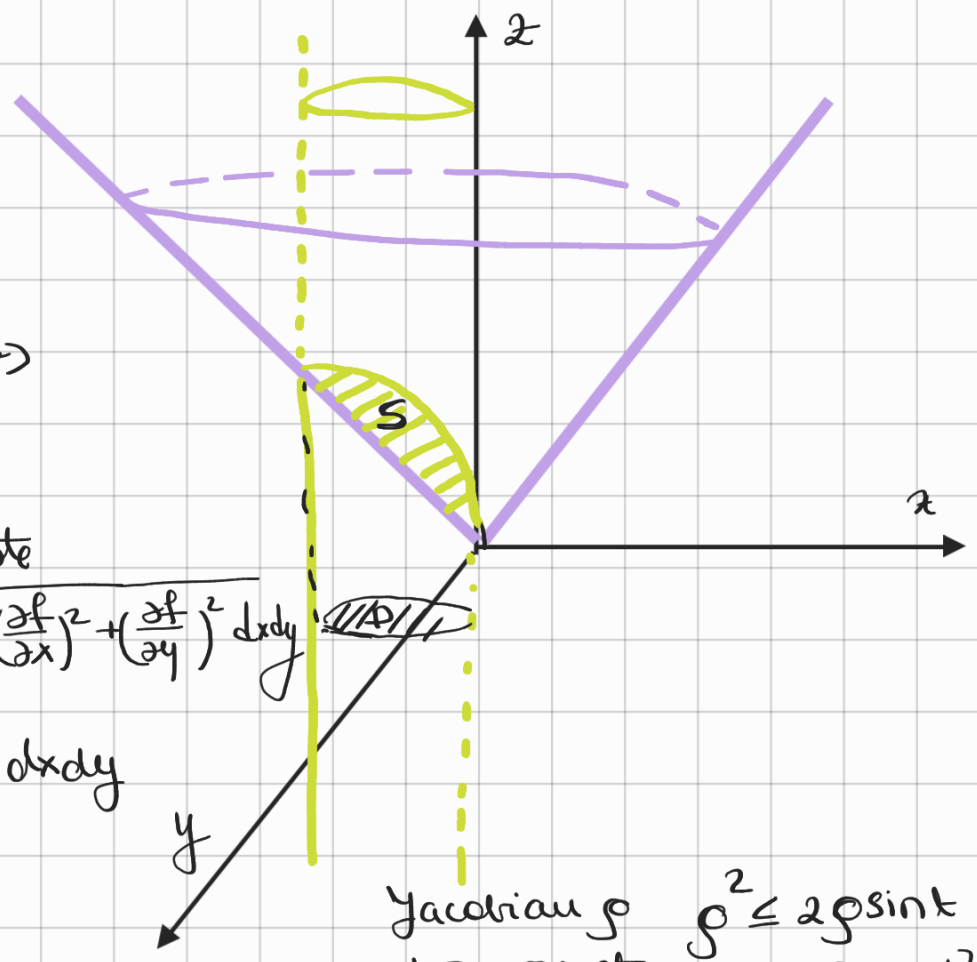
$$\Rightarrow \mathcal{A} = \int_0^{2\pi} dt \int_0^{1/\sqrt{2}} \sqrt{1 + 4\rho^2} \cdot \rho \, d\rho = \int_0^{2\pi} \frac{3^{3/2} - 1}{12} dt = \frac{3^{3/2} - 1}{12} \cdot t \Big|_0^{2\pi}$$

$$= \frac{3^{3/2} - 1}{6} \cdot \pi$$

cau

Să se determine aria porțiunii de pe suprafața $\sqrt{x^2+y^2}-z=0$, situată în interiorul suprafeței $x^2+y^2-2y=0$. Reprezentare grafică.

↳ cilindru



$z=\sqrt{x^2+y^2}=f(x,y)$ ⇒

$\left.\begin{aligned}\frac{\partial f}{\partial x}&=\frac{x}{\sqrt{x^2+y^2}}\\ \frac{\partial f}{\partial y}&=\frac{y}{\sqrt{x^2+y^2}}\end{aligned}\right\}\Rightarrow$

Aria suprafeței este

$$\mathcal{A}=\int\limits_{\Sigma}\int d\sigma=\iint\limits_{D}\sqrt{1+\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2}\,dxdy$$

$$=\iint\limits_{D}\sqrt{1+\frac{x^2}{x^2+y^2}+\frac{y^2}{x^2+y^2}}\,dxdy$$

$$=\sqrt{2}\iint\limits_{D}dxdy,$$

unde $D=\{(x,y)\mid x^2+y^2\le 2y\}$

Jacobian ρ: $\begin{cases}x=\rho\cos t\\ y=\rho\sin t\end{cases}$ $\rho^2\le 2\rho\sin t \Rightarrow \rho\in[0,2\sin t]$, $t\in[0,\pi]$

$$\iint\limits_{D}dxdy=\int\limits_{0}^{\pi/2}dt\int\limits_{0}^{2\sin t}\rho\,d\rho=\int\limits_{0}^{\pi}\frac{\rho^2}{2}\Big|_0^{2\sin t}dt=\int\limits_{0}^{\pi}\frac{4\sin^2 t}{2}dt=$$

$$=\int\limits_{0}^{\pi}2\sin^2 t\,dt=\int\limits_{0}^{\pi}(1-\cos 2t)\,dt=\left(t-\frac{1}{2}\sin 2t\right)\Big|_0^{\pi}=$$

$$=\pi-0+0-0=\pi\Rightarrow$$

$$\mathcal{A}=\sqrt{2}\pi$$